

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT (2 Questions)

Course Code:	ITA0506	Course Title:	COMPUTER VISION
CO Assessed:	CO2 – Manipulation (BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
CO5 Statement:	Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BL3)		
Student Name:	Y Sravan Kumar	Reg. No.:	192472289
Section / Batch:	CSE AI Slot D	Date:	1/8/26

Code of Conduct

I, Y Sravan Kumar(192472289) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Y Sravan Kumar

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus Area	Q. Nos.	Marks
Frequency Domain Image Enhancement	Periodic noise removal, Fourier Transform, notch filtering, quality inspection	1	10
Spatial Filtering & Edge-Preserving Enhancement	Bilateral filtering, edge preservation, autonomous vehicle navigation	2	10
GRAND TOTAL:			20 Marks

Annexure A – Scenario Based Assessment

Scenario 1: Manufacturing Quality Inspection

An automated quality inspection system captures images of metal components. Electrical interference introduces repetitive horizontal stripes across the images, making defect detection unreliable.

- (a) Interpret the scenario and identify the type of noise present.**
- (b) Explain how this degradation affects inspection accuracy.**
- (c) Recommend an appropriate enhancement technique to remove the interference.**
- (d) Justify why your selected approach is more suitable than spatial-domain methods.**
- (e) Explain your answer in a logical sequence.**

Scenario 2: Autonomous Vehicle Navigation

A self-driving car captures road images during heavy rain. Although noise is present, the lane markings and road edges must remain sharp for safe navigation.

- (a) Interpret the scenario and explain the challenge involved.**
- (b) Identify the importance of preserving edges while removing noise.**
- (c) Recommend a suitable filtering technique.**
- (d) Justify why your selected filter preserves important image details.**
- (e) Present your explanation clearly and systematically.**

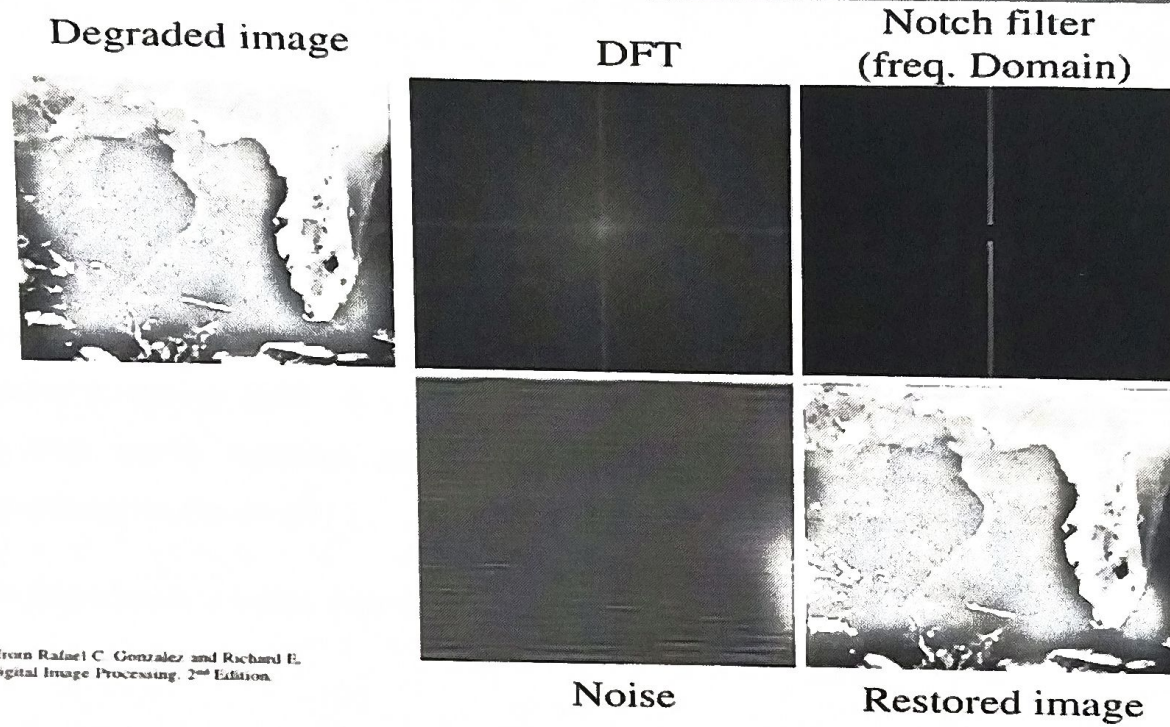
Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks	
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario		
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues		
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts		
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided		
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		

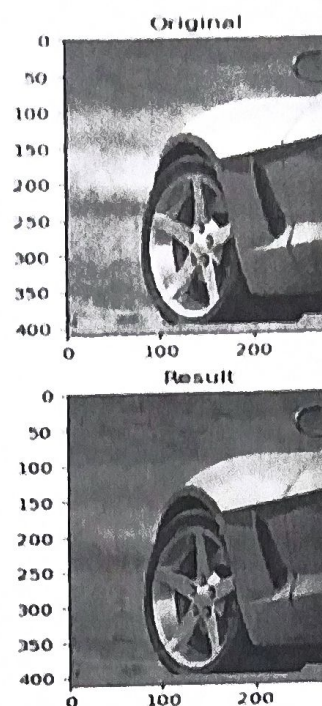
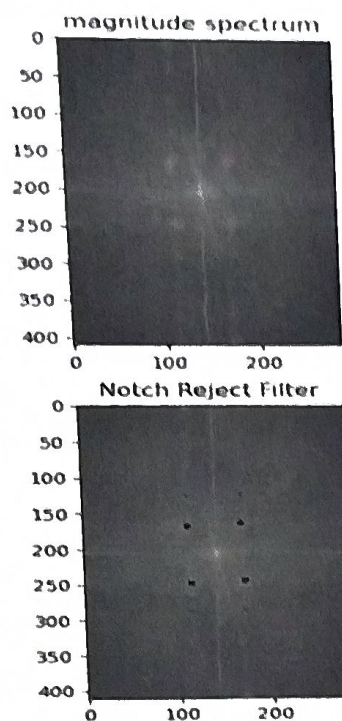
Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Scenario 1: Manufacturing Quality Inspection

Notch Reject Filter:



(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.



(a) Interpret the scenario and identify the type of noise present.

The manufacturing quality inspection system captures images of metal components to detect defects such as cracks, scratches, dents, or surface imperfections. During image acquisition, electrical interference from nearby machines, power lines, or electronic equipment introduces repetitive horizontal stripes across the image.

This type of degradation is called Periodic Noise because it appears at regular intervals as repeated horizontal or vertical patterns.

Characteristics of Periodic Noise:

- Appears as repeated horizontal or vertical lines.
- Caused by electrical or electromagnetic interference.
- Has a regular, predictable pattern.
- Clearly visible in both the image and its frequency spectrum.

Answer:

Type of Noise: Periodic Noise (Electrical Interference).

(b) Explain how this degradation affects inspection accuracy.

Periodic noise significantly reduces the performance of an automated inspection system because it hides important image details.

Effects on inspection:

- Masks small defects such as cracks, scratches, and pits.
- Reduces image contrast and sharpness.
- Produces false defect detections (false positives).
- Misses actual defects (false negatives).
- Reduces the accuracy of edge detection and segmentation.
- Decreases the reliability of quality control.
- Can result in defective products being accepted or good products being rejected.

Example:

A small crack on a metal surface may overlap with the horizontal stripes, making it difficult for the computer vision system to distinguish between the crack and the noise.

(c) Recommend an appropriate enhancement technique to remove the interference.

The most suitable enhancement technique is Frequency Domain Filtering using a Notch Reject Filter.

Working Procedure:

1. Capture the noisy image.
2. Apply the Fast Fourier Transform (FFT) to convert the image into the frequency domain.
3. Identify the bright frequency peaks produced by periodic noise.
4. Apply a Notch Reject Filter to remove only those unwanted frequencies.
5. Apply the Inverse FFT (IFFT) to reconstruct the cleaned image.

Advantages:

- Removes periodic noise effectively.
- Preserves edges and fine details.
- Improves image quality for accurate defect detection.

(d) Justify why your selected approach is more suitable than spatial-domain methods.

Periodic noise is much easier to identify in the frequency domain than in the spatial domain.

Frequency Domain (NotchFilter)	Spatial Domain Filters
Removes only periodic frequencies	Filters the whole image
Preserves edges and details	May blur edges and fine features
Highly effective for stripe noise	Not effective for periodic interference
Improves defect detection accuracy	Can reduce image quality

Justification:

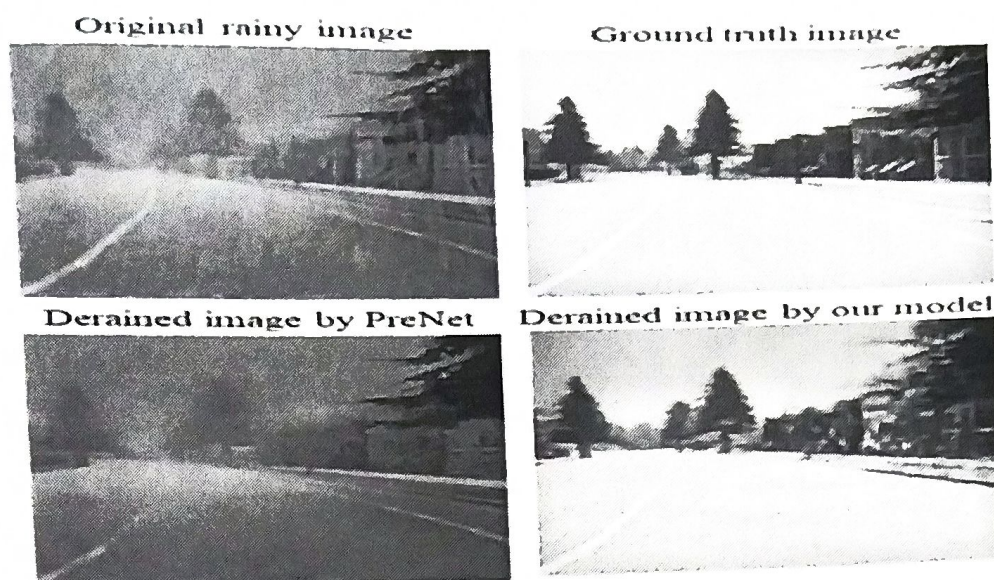
- The FFT separates the image into different frequency components.
- Periodic noise appears as distinct peaks in the frequency spectrum.
- A Notch Reject Filter targets only these peaks without affecting useful image information.
- Spatial filters such as the Mean Filter, Median Filter, or Gaussian Filter are better suited for random noises (e.g., Gaussian or Salt-and-Pepper noise), but they cannot effectively remove repetitive stripe patterns and often blur the image.

Conclusion:

A Notch Reject Filter in the Frequency Domain is the best choice because it selectively removes periodic interference while preserving important defect details, leading to higher inspection accuracy.

Scenario 2: Autonomous Vehicle Navigation





(a) Interpret the scenario and explain the challenge involved.

A self-driving car uses cameras to capture road images for detecting lane markings, road edges, vehicles, pedestrians, and traffic signs. During heavy rain, the captured images contain noise caused by rain droplets, water splashes, and poor visibility.

The main challenge is to remove the noise without blurring important road features, especially the lane markings and road edges, because these features are essential for safe navigation.

Challenges:

- Rain introduces noise into the image.
- Reduced visibility makes object detection difficult.
- Lane markings become less clear.
- The system must clean the image while preserving important details.

(b) Identify the importance of preserving edges while removing noise.

Edges represent the boundaries of objects in an image. In autonomous driving, preserving edges is critical because they help the vehicle identify:

- Lane markings.
- Road boundaries.

- Vehicles.
- Pedestrians.
- Traffic signs and obstacles.

If edges become blurred:

- Lane detection becomes inaccurate.
- Object detection performance decreases.
- Navigation decisions become unreliable.
- The risk of accidents increases.

Importance of edge preservation:

- Maintains accurate lane detection.
- Improves object recognition.
- Supports safe steering decisions.
- Enhances overall navigation accuracy.

(c) Recommend a suitable filtering technique.

The most suitable filtering technique is the Median Filter.

Working Procedure

1. Capture the noisy road image.
2. Select a small neighborhood (e.g., 3×3 or 5×5 window).
3. Arrange the pixel values in ascending order.
4. Replace the center pixel with the median value.
5. Repeat for all pixels in the image.

Advantages

- Removes noise effectively.

5. The filter preserves edges while reducing unwanted noise.
6. The enhanced image improves lane detection, object recognition, and safe vehicle navigation.

Justification:

The Median Filter is preferred because it removes noise without blurring important edges. It preserves lane markings and road boundaries, improving lane detection and ensuring accurate and safe navigation for autonomous vehicles.

Conclusion:

The Median Filter effectively removes rain noise while preserving lane markings and road edges, improving image quality and ensuring accurate and safe autonomous vehicle navigation.